Plant Disease Detection using deep learning: CNN Multiclass Classification

2025

THESIS SUBMITTED TO

Symbiosis Institute of Geoinformatics

FOR PARTIAL FULFILLMENT OF THE M. Sc. DEGREE

By

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Batch of 2024-2026

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Acknowledgement

I express my deepest gratitude to our director, Dr. T. P. Singh, of Symbiosis Institute of Geoinformatics (SIG), for allowing us to conduct a two-month research project. I am expressing my most profound and sincere gratitude towards Symbiosis Institute of Geoinformatics (SIG) and Mrs. Neeta Nandgude for their invaluable contributions to this project. My most profound appreciation goes to my internal guide, Mrs. Neeta Nandgude, for their continuous support, constant guidance, and constructive feedback throughout the project. I also sincerely thank the Symbiosis Institute of Geoinformatics (SIG) for providing the facilities and resources necessary for this project's completion.

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Abbreviation List

CNN: Convolutional Neural Network

XGBoost: Extreme Gradient Boosting

TF: TensorFlow, Keras

Performance Metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Preface

This paper presents work on image based automated system for plant disease detection from leaf images. The method is based on a hybrid machine learning model that includes Convolutional Neural Network as feature extractor due to its excellent performance in this process and Extreme Gradient Boosting as classifier because of the effectiveness of these classifiers. The goal is to develop a scalable, accurate and user-friendly approach to aid farmers and agriculturalists in early detection of plant diseases. This will reduce crop damage.

Introduction

Plant diseases are the main causes which result in a drop of the output and, in turn, inflicting the loss of money in the field of agriculture. Precisely identifying a disease is a tough and challenging task that also requires some skills. Besides, the disease or its symptoms like discolored spots or streaks on the leaves of the plant can be visible. Most of the time, pathogens that consist of fungi, bacteria, and viruses are the main causes of plants diseases. The symptoms symptoms and signs are quite a few that vary depending on the cause or etiology of the plant disease.

The neural networks have been an extremely successful implementation in various and diverse domains, as examples of end-to-end learning.The neurons in a neural network are mathematical units that perform a function for the numerical inputs taken from the incoming edges and give a numerical result as an outgoing edge. The CNN may also have its applications in the agricultural field that includes the identification of diseases as well as to the diseased area for quantification.

'The New Plant Disease Detection Dataset' has amassed thousands of plant images and has made the collected data free and accessible to anyone willing to use it. The dataset is already processed. Besides the end-to-end CNN method, the hybrid model that integrates a feature extractor that was pre-trained with an XGBOOST classifier is another option for identifying plant diseases. Such a method is highly effective when dealing complicated high-dimensional data such as image features, and solving the problem of class imbalance in the dataset by attributing weights to each class can be a further tuning of the model.

Problem Statement

The primary issue that the current research desires to resolve is building a method that can accurately and reliably detect plant diseases just by pictures. A technology like this must be able to recognize not only the differences of the diseases that have been given, but also the healthy plants regardless of the various complexities of nature or farm setting. These complications may include different lights, dirty or disordered backgrounds and the quality of the pictures which may not be consistent. The point, in the end, is to have a usable and flexible model that can work well under the different conditions.

Objectives

Building and implementing a high-performance deep learning model: It basically means creating a deep learning model that can visually identify the normal diseases of common plants by the pictures of their leaves.

Delegating the model to be proficient and dependable: The model will be trained using a big and diverse dataset of plant images, and its performance will be validated thoroughly to ensure its trustworthiness.

Developing a user-friendly and practical application: An application that is functional and simple enough for users to get quick disease diagnoses and suggestions from the trained model, will be designed.

Literature Review

Most reviewed studies tell us that combining Convolutional Neural Networks (CNNs) with XGBoost is a very good idea when we are dealing with multiclass classification problems across various domains such as agriculture, medical imaging, bioinformatics, and signal recognition.

In the field of agriculture, different scientific works (Hassan et al., 2021; Inderscience, 2022; SCIRP, 2024) reveal that using features extracted by CNN and then applying them to XGBoost results in very correct researches (most of the time more than 95%) related to multicrop and plant disease datasets. The said combination not only achieves better performance in terms of accuracy but also is able to gain higher robustness, especially against those situations of data that have very little samples of certain classes or have been contaminated by noise.

Machine learning methods like CNN + XGBoost hybrids have already been employed to automatic pneumonia, liver cancer, lymphoma, and COVID-19 infection detection. The effects of models combining these techniques are always better than those of the only CNN approach (Zivković et al., 2022; Hamdi et al., 2023; JODAC, 2024). The authors of theses research deliver truthably accuracy rates accession from 87% to 97%, thereby confirming the effectiveness of the pipeline in the most documented and hard-to-treat diseases/areas setting of healthcare.

Moreover, the application of the CNN + XGBoost model in the biomedical field has been extensively explored for the prediction of protein localization combined with the detection of the correct cell type (Frontiers, 2018; XgCPred, 2024); while in the field of signals and emotions as well as in traffic sign recognition, the model is highly cited (Khan et al., 2022; Elsevier, 2022). So, in all these various works, it is mentioned that the joint scheme (combination) produces the highest macro-F1 and best recall for the minority class, thus making it an efficient solution to the particular multiclasses that suffer from the distribution of the classes qualities.

Besides that, one main aspect of these and other similar papers (MDPI, 2021; Raju et al., 2024) is that the proposed solutions of hybrid architectures of the kind CNN + XGBoost can become more performing after further optimizations realized by Adaptive Particle Swarm Optimization (APSO) or by merging another architecture, e.g., Vision Transformers.

Review Papers

Sr.no	**Author(s) / Year**	**Dataset / Domain**	**Method (CNN + XGBoost)**	**Result / Accuracy**
1.	Hassan et al., 2021	PlantVillage	Shallow CNN features + XGBoost	Improved over CNN, ~97%
2.	Inderscience, 2022 (ConRXG)	PlantVillage (transfer learning)	ResNet50 features + XGBoost	High accuracy (~98%)
3.	JISEM, 2022	Palm tree leaves	CNN features + XGBoost	~97% accuracy
4.	SCIRP, 2024	Cocoa leaf diseases	CNN feature extractor + XGBoost	Robust multiclass performance
5.	Kaggle Study, 2023	Tomato leaf dataset	Hybrid CNN + XGBoost	Higher generalization vs CNN
6.	Thakur et al., 2023	PlantVillage subset	Lightweight CNN + SVM vs XGBoost	Hybrid outperformed baseline
7.	Zivković et al., 2022	Chest X-ray (COVID-19)	CNN + XGBoost classifier	Reliable detection, >95%
8.	Hamdi et al., 2023	Lymphoma images	CNN feature fusion + XGBoost	Better than CNN-only
9.	PubMed, 2023	Pneumonia X-ray images	CNN features + XGBoost	~87% accuracy
10.	JODAC, 2024	Liver cancer detection	CNN feature extraction + XGBoost	~95.2% accuracy

11.	AIJR, 2022	General medical images	CNN features → XGBoost	Hybrid better than CNN
12.	Frontiers, 2018	Protein localization	CNN features + XGBoost	Improved subcellular localization
13.	XgCPred, 2024	scRNA-seq data	Image conversion + CNN features + XGBoost	High classification accuracy
14.	Mavaie et al., 2023	Biomedical low-resource data	CNN features + XGBoost	Works well on small datasets
15.	Khan et al., 2022	EEG spectrograms	CNN features + XGBoost	Emotion recognition >92%
16.	Roumeliotis et al., 2025	Crop datasets	Multimodal CNN + XGBoost	Improved crop disease classification
17.	Elsevier, 2022	Traffic signs	CNN extractor + XGBoost	Boosted recognition accuracy
18	MDPI, 2021	General image datasets	CNN features + XGBoost + APSO tuning	Higher than CNN baseline
19.	Raju et al., 2024	Vision Transformers vs CNN	CNN + XGBoost vs ViT hybrids	Competitive performance
20.	Conf. Paper, 2022	Small datasets (imbalanced)	CNN features + XGBoost	Better generalization under imbalance

Hassan et al. (2021) – Applied shallow CNN models on the PlantVillage dataset for multiclass plant disease classification. Features were extracted and classified using XGBoost, improving accuracy (~97%) compared to pure CNN.

Inderscience (ConRXG, 2022) – Used transfer learning with ResNet50 to extract leaf features and XGBoost for final classification. Achieved ~98% accuracy, showing that the hybrid approach generalized well across 58 plant classes.

JISEM (2022) – Proposed a CNN + XGBoost hybrid for palm tree leaf disease detection. The model achieved ~97% accuracy, demonstrating fast and reliable classification across multiple palm diseases.

SCIRP (2024) – Focused on cocoa leaf disease detection using CNN features with XGBoost classification. Results showed high robustness in distinguishing between multiple leaf disease classes.

Thakur et al. (2023) – Developed a lightweight CNN feature extractor and compared XGBoost with SVM for multiclass plant disease detection. The CNN + XGBoost combination outperformed CNN alone in accuracy and generalization.

Kaggle Tomato Study (2023) – Used CNN pretrained models to extract features from tomato leaf images and trained XGBoost for classification. The hybrid model improved multiclass accuracy under varied real-world conditions.

Zivković et al. (2022) – Applied CNN + XGBoost on chest X-ray datasets for multiclass COVID-19 detection. The hybrid system reached >95% accuracy and proved more reliable than standalone CNN softmax classifiers.

Hamdi et al. (2023) – Used CNN feature fusion and XGBoost for lymphoma subtype classification. The model showed stronger performance in multiclass settings by balancing precision and recall across categories.

PubMed Study (2023) – Built a CNN + XGBoost pipeline to classify pneumonia X-rays into viral, bacterial, and normal cases. Achieved ~87% accuracy, demonstrating the feasibility of hybrids for medical imaging multiclass tasks.

ODAC (2024) – Proposed a CNN + XGBoost hybrid for liver cancer detection. Reported ~95.2% accuracy, highlighting the hybrid's potential in critical healthcare diagnosis tasks.

AIJR (2022) – Used CNN embeddings as input to XGBoost for general medical image classification. The hybrid achieved higher multiclass accuracy compared to CNN-only architectures.

Frontiers in Genetics (2018) – Applied CNN + XGBoost for protein subcellular localization across multiple classes. The hybrid improved prediction accuracy compared to standalone CNNs, making it useful for bioinformatics.

XgCPred (2024, ScienceDirect) – Converted scRNA-seq data into image-like representations, used CNN for feature extraction, and XGBoost for cell-type classification. Achieved high multiclass accuracy on complex genomic datasets.

Mavaie et al. (2023, BMC Bioinformatics) – Investigated CNN + XGBoost for biomedical images with limited training samples. Demonstrated the hybrid's ability to maintain strong multiclass classification in low-resource conditions.

Khan et al. (2022, Sci. Reports) – Built a CNN + XGBoost model for emotion recognition using EEG spectrograms. The hybrid system achieved >92% multiclass accuracy, outperforming standalone CNNs.

Roumeliotis et al. (2025) – Applied CNN + XGBoost for multicrop plant disease classification. The hybrid reduced misclassification rates and improved precision across crop classes compared to CNN alone.

Elsevier Study (2022) – Used CNN features with XGBoost for traffic sign recognition. The hybrid pipeline improved classification rates in multiclass traffic sign datasets, proving useful for autonomous driving.

MDPI (2021, APSO optimization) – Proposed CNN feature extraction with XGBoost tuned using Adaptive Particle Swarm Optimization (APSO). Showed improved multiclass accuracy compared to vanilla CNN classifiers.

Raju et al. (2024, Sci. Reports) – Compared CNN + XGBoost hybrids with Vision Transformer + XGBoost on image datasets. Found CNN + XGBoost to be competitive, particularly on smaller multiclass datasets.

Conference Paper (2022) – Focused on multiclass classification of small, imbalanced datasets. CNN features combined with XGBoost improved macro-F1 and minority-class recall compared to CNN softmax classifiers

Methodology

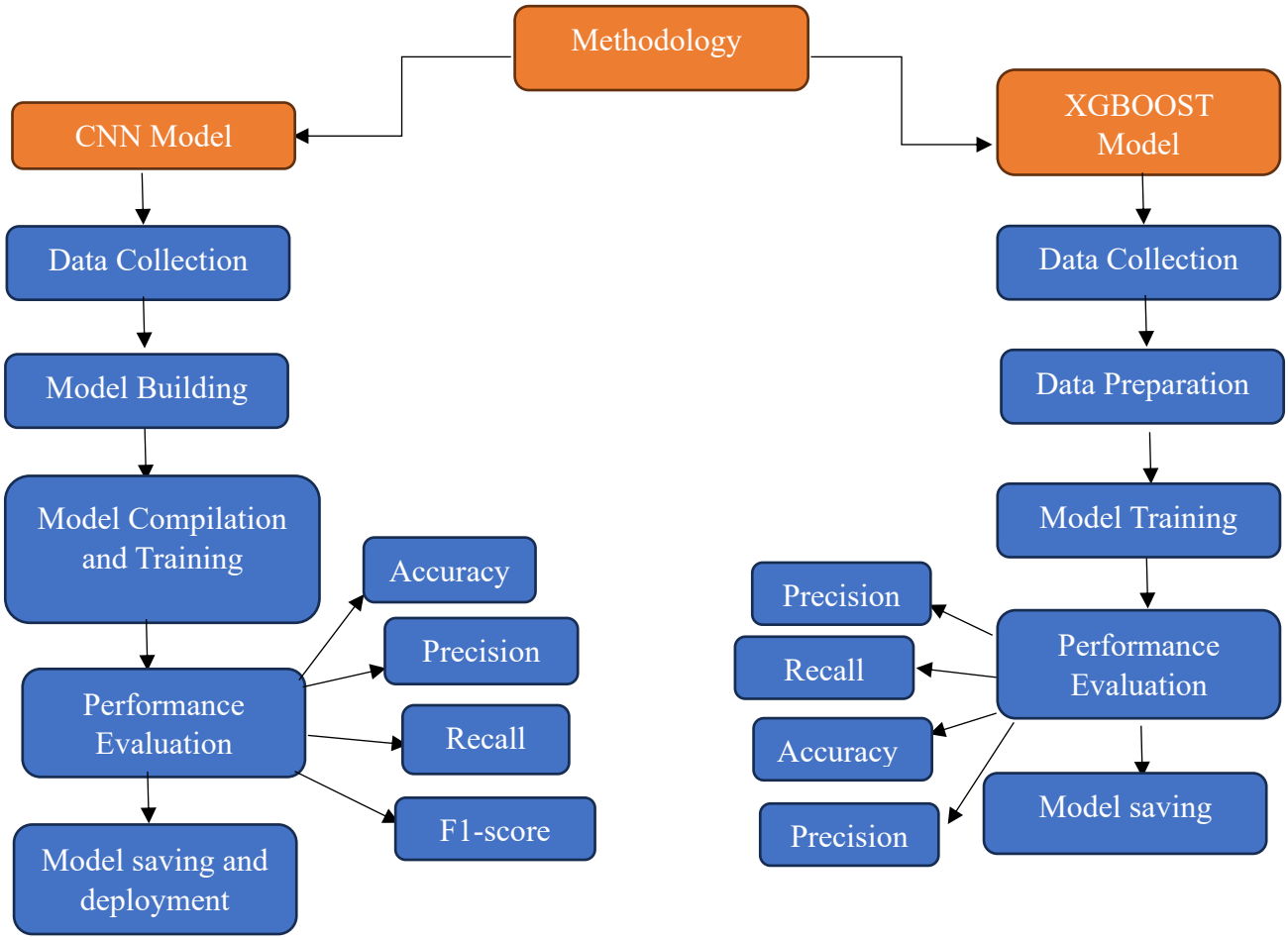


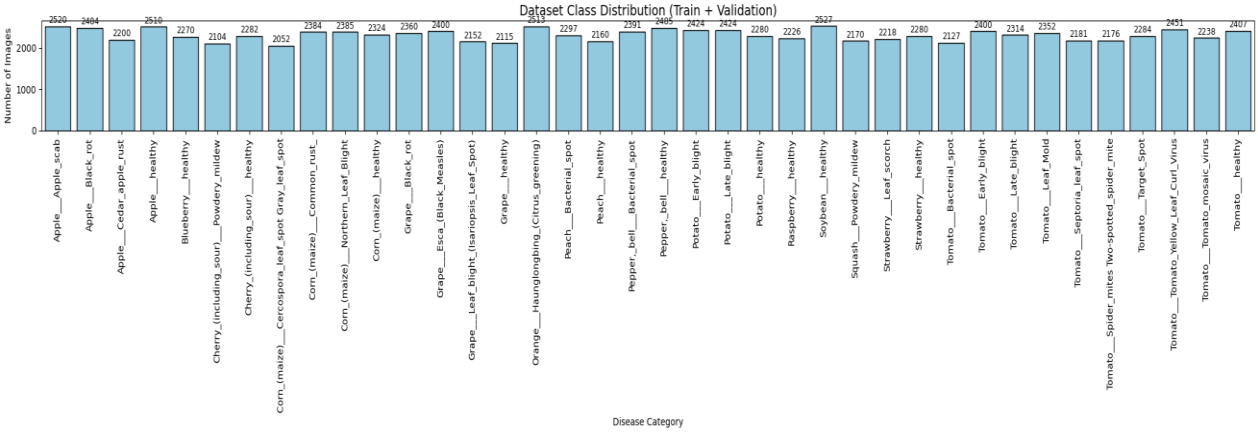
Figure 1: Workflow of methodology

Data Collection

As shown in Figure 1,The foundation of this research is the PlantVillage dataset, a public repository of images of various plants, both healthy and diseased. The dataset was formally partitioned into three distinct subsets to ensure an unbiased evaluation:

- **Train:** A huge set of 70,295 images for the model to learn from.
- **Validation:** 17,572 images to help us fine-tune the model as it trained.
- **Test:** 33 images for testing part.

Figure 2 : Some plant dataset images



Dataset Distribution (Train+Valid)

Model Selection:

Model Architecture: CNN

1. Model Building: A sequential CNN model is constructed with multiple Conv2D and MaxPool2D layers, interspersed with Dropout layers to prevent overfitting. The model's architecture also includes Flatten and Dense layers, with the final output layer using a 'softmax' activation function for classification.
2. Model Compilation: The model is compiled using the Adam optimizer with a learning rate of 0.0001 and 'categorical_crossentropy' as the loss function.
3. Model Training: The model is trained on the preprocessed training and validation datasets for 10 epochs.
4. Evaluation: The trained model's performance is evaluated using accuracy and loss metrics on both the training and validation sets. Additional metrics like a classification report and a confusion matrix are generated to further assess performance.

5. Model Saving and Prediction: The trained model is saved as "trained_model.keras" and can be loaded for future predictions on new images.
6. Each of the five convolutional blocks constitutes the CNN architecture.It include 5 layers:

   - Block 1: Two Conv layers (32 filters) + MaxPooling – the layer was designed to deal with basic features of the image such as edges.
   - Block 2: Two Conv layers (64 filters) + MaxPooling – the layer learns textures and shapes of the objects.
   - Block 3: Two Conv layers (128 filters) + MaxPooling – the layer identifies the infection patterns.
   - Block 4: Two Conv layers (256 filters) + MaxPooling – the layer isolates the smallest details.
   - Block 5: Two Conv layers (512 filters) + MaxPooling – the layer gets the most abstract representations of the input.

Model Evaluation

```
[59] print(classification_report(Y_true,predicted_categories,target_names=class_name))
```

```
...                                                   precision    recall  f1-score   support

                                 Apple___Apple_scab       0.97      0.93      0.95       504
                                  Apple___Black_rot       0.89      0.98      0.93       497
                            Apple___Cedar_apple_rust       0.93      0.99      0.96       440
                                     Apple___healthy       0.97      0.93      0.95       502
                                Blueberry___healthy       0.93      0.99      0.96       454
             Cherry_(including_sour)___Powdery_mildew       1.00      0.93      0.96       421
                    Cherry_(including_sour)___healthy       0.95      1.00      0.97       456
     Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot       0.98      0.87      0.92       410
                          Corn_(maize)___Common_rust_       1.00      0.99      1.00       477
                  Corn_(maize)___Northern_Leaf_Blight       0.90      0.99      0.94       477
                           Corn_(maize)___healthy       1.00      1.00      1.00       465
                                 Grape___Black_rot       0.99      0.96      0.98       472
                       Grape___Esca_(Black_Measles)       0.99      0.99      0.99       480
            Grape___Leaf_blight_(Isariopsis_Leaf_Spot)       0.98      1.00      0.99       430
                                  Grape___healthy       0.96      1.00      0.98       423
             Orange___Haunglongbing_(Citrus_greening)       0.99      0.98      0.98       503
                           Peach___Bacterial_spot       0.96      0.94      0.95       459
                                   Peach___healthy       0.94      1.00      0.97       432
                     Pepper,_bell___Bacterial_spot       0.98      0.84      0.90       478
                            Pepper,_bell___healthy       0.90      0.97      0.94       497
                             Potato___Early_blight       0.90      0.99      0.94       485
                              Potato___Late_blight       0.80      0.99      0.88       485
                                  Potato___healthy       0.95      0.96      0.96       456
    ...
                                          accuracy                           0.95     17572
                                         macro avg       0.95      0.95      0.95     17572
                                      weighted avg       0.95      0.95      0.95     17572
```

Figure 3: Classification Report CNN(Precision, Recall, F1-score

As shown in Figure 3, the evaluation includes:

Accuracy	95%
Precision	95%
Recall	95%
F1-score	95%

Table1:Classification of CNN

The CNN Model trained and pass 10 epochs and it gives **95% of accuracy**.

2.XGBOOST Model:

1.Handling Class Imbalance: To address inbalanced classes,this utilizes the compute weight function from sklearn.utils.

2.Model Training: An XGBClassifier is instantiated with specific hyperparameters such as n_estimators=600 and learning_rate=0.05. The model is then trained using the extracted features and the calculated class weights.

3.Evaluation: The model is evaluated on a validation set, and a classification report and confusion matrix are generated to measure its performance.

XGBOOST model were meticulously tuned to optimize performance, with key parameters including:

Ntree=600

Maxdepth=8

Learningrate=0.05

Subsample=0.9

Colsamplebytree=0.9

```
...   Validation Accuracy: 0.9763
      Validation Classification Report:
                                                        precision    recall  f1-score   support

                                   Apple___Apple_scab     0.9819    0.9683    0.9750       504
                                    Apple___Black_rot     0.9860    0.9899    0.9880       497
                             Apple___Cedar_apple_rust     0.9932    0.9886    0.9909       440
                                      Apple___healthy     0.9783    0.9861    0.9821       502
                                 Blueberry___healthy     0.9847    0.9934    0.9890       454
            Cherry_(including_sour)___Powdery_mildew     0.9976    1.0000    0.9988       421
                   Cherry_(including_sour)___healthy     0.9913    0.9978    0.9945       456
      Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot     0.9713    0.9073    0.9382       410
                         Corn_(maize)___Common_rust_     0.9958    1.0000    0.9979       477
                 Corn_(maize)___Northern_Leaf_Blight     0.9264    0.9769    0.9510       477
                              Corn_(maize)___healthy     0.9979    1.0000    0.9989       465
                                   Grape___Black_rot     0.9769    0.9852    0.9810       472
                        Grape___Esca_(Black_Measles)     0.9853    0.9792    0.9822       480
           Grape___Leaf_blight_(Isariopsis_Leaf_Spot)     1.0000    0.9953    0.9977       430
                                      Grape___healthy     0.9976    0.9976    0.9976       423
             Orange___Haunglongbing_(Citrus_greening)     1.0000    0.9901    0.9950       503
                              Peach___Bacterial_spot     0.9912    0.9782    0.9846       459
                                      Peach___healthy     0.9772    0.9907    0.9839       432
                       Pepper,_bell___Bacterial_spot     0.9834    0.9916    0.9875       478
                              Pepper,_bell___healthy     0.9820    0.9859    0.9839       497
                               Potato___Early_blight     0.9897    0.9938    0.9918       485
      ...
                                             accuracy                         0.9763     17572
                                            macro avg     0.9764    0.9761    0.9761     17572
                                         weighted avg     0.9764    0.9763    0.9762     17572
```

Figure 4: Classification report XGBOOST

As shown in Figure 2,the classification report of XGBOOST include :

Accuracy	97.63%
Precision	97.64%
Recall	97.61%
F1-score	97.61

Table 2:Classification of XGBOOS

Model Deployment

1. Open the App: Simply visit the web page where the app is available. Being made with Streamlit, the app is user-friendly and neat.

2. Upload a Picture: There is an area to upload a pretty clear picture of your plant's leaf.

3. Get a Prediction: By clicking the "Predict" button, the app will be activated. While the model is working, a little loading message will be visible.

4. See the Results: The app will, in a short time, let you know the exact disease it suspects the plant to have.

5. Read the Recommendations: The app does not only provide the diagnosis, but it also gives advice on what fertilizer to use, how to take care of the plant, and how to prevent the disease from coming back. The information concerning plant disease fertilizers, care, tips, symptoms and preventive measures is provided in one file which reads the data and recommends the tips for diseased plants.

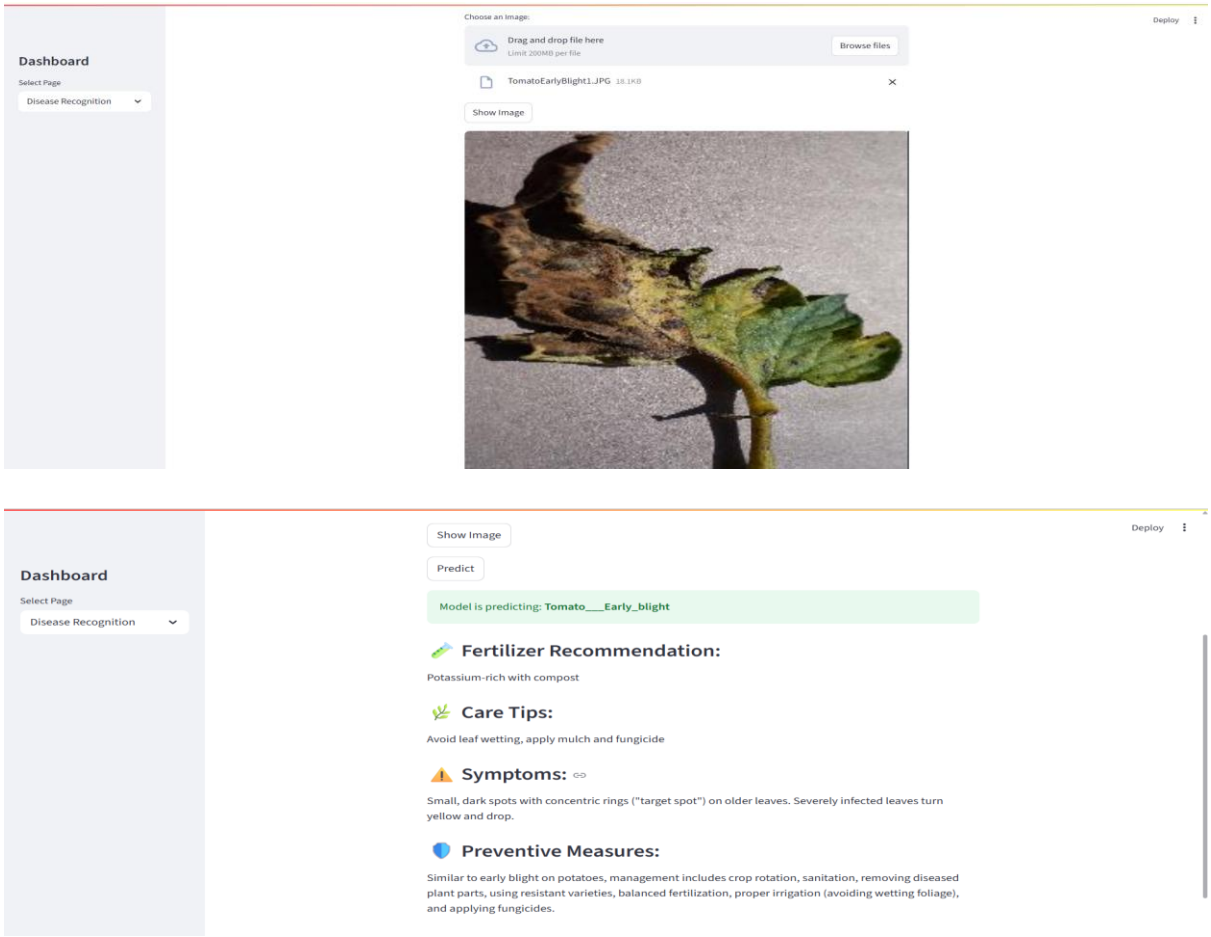


Figure 5: Prediction result and recommendations

Results

1.Visualization of Accuracy of CNN Model

```
epochs=[i for i in range(1,11)]
plt.plot(epochs,training_history.history['accuracy'],color='red',label='Training Accuracy')
plt.plot(epochs,training_history.history['val_accuracy'],color='blue',label='Validation Accuracy')
plt.xlabel("No of Epochs")
plt.ylabel("Accuracy Result")
plt.title("Visualization of Accuracy Result")
plt.legend()
plt.show()
```

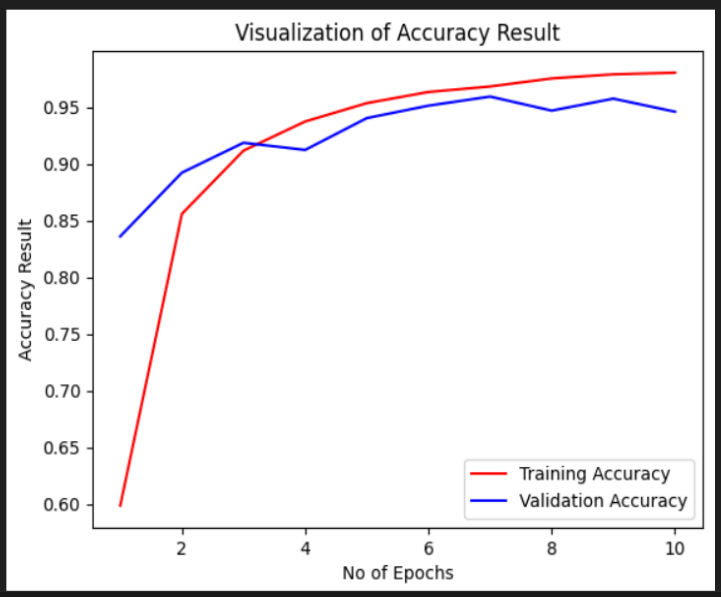


Figure 6: Visualization of CNN Model accuracy results

As shown in Figure 6, The Training Accuracy keeps on getting better with each epoch, eventually, it gets very close to 0.95 at the end of the training.The Validation Accuracy gets better at the start,however, it seems to be leveling out and slightly dropping at the later epochs, which could be a sign of overfitting.The difference between Training Accuracy and Validation Accuracy gets bigger over time, which is typical for machine learning situations, indicating the model is losing its ability to generalize.

2.Confusion Matrix CNN

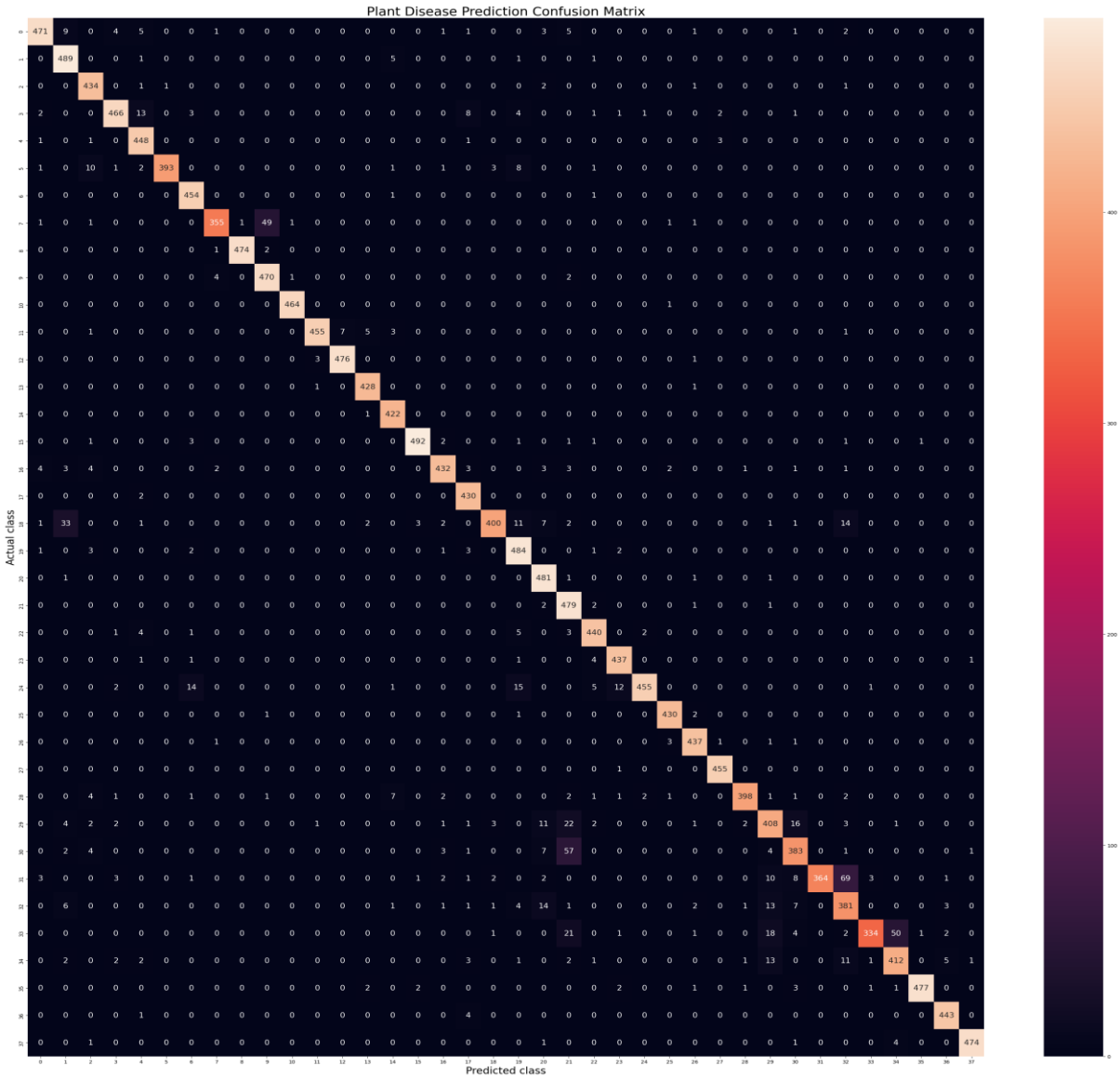


Figure 7: Confusion Matrix Of CNN

1.Diagonal Elements (True Positives)

As shown in Figure 7, The diagonal elements indicate the number of truly positive (the category predicted matches the actual category). Most of the numbers on the diagonal are pretty big, which means that the model is good at identifying plant diseases correctly. For instance:

- Class 0: 472 times correct prediction made by the model
- Class 1: 489 times the model made accurate prediction
- Class 2: 492 times the model correctly made the prediction, and so on

2.Off-Diagonal Elements (Misclassifications):

Off-diagonal elements reflect those misclassifications, whereby the class predicted does not equal the class observed**.**Some off-diagonal numbers are quite sizable for certain classes, that may be the points that the model is not working well in those areas:To be more specific, Class 6 (with 57 misclassifications), and Class 10 (with 22 misclassifications), that might imply these classes being confused with one another and some others.Besides, Class 13 is not an exception that has a large number of misclassifications, 33 instances that were distributed into other classes.

3.Overall Performance:

Most classes show high numbers of correct predictions, suggesting that the model is able to predict plant diseases with good accuracy. The model has shown strong performance overall, with correct predictions mostly concentrated along the diagonal of the confusion matrix. This suggests that the model is able to distinguish between plant diseases effectively.

3.Confusion Matrix XGBOOST

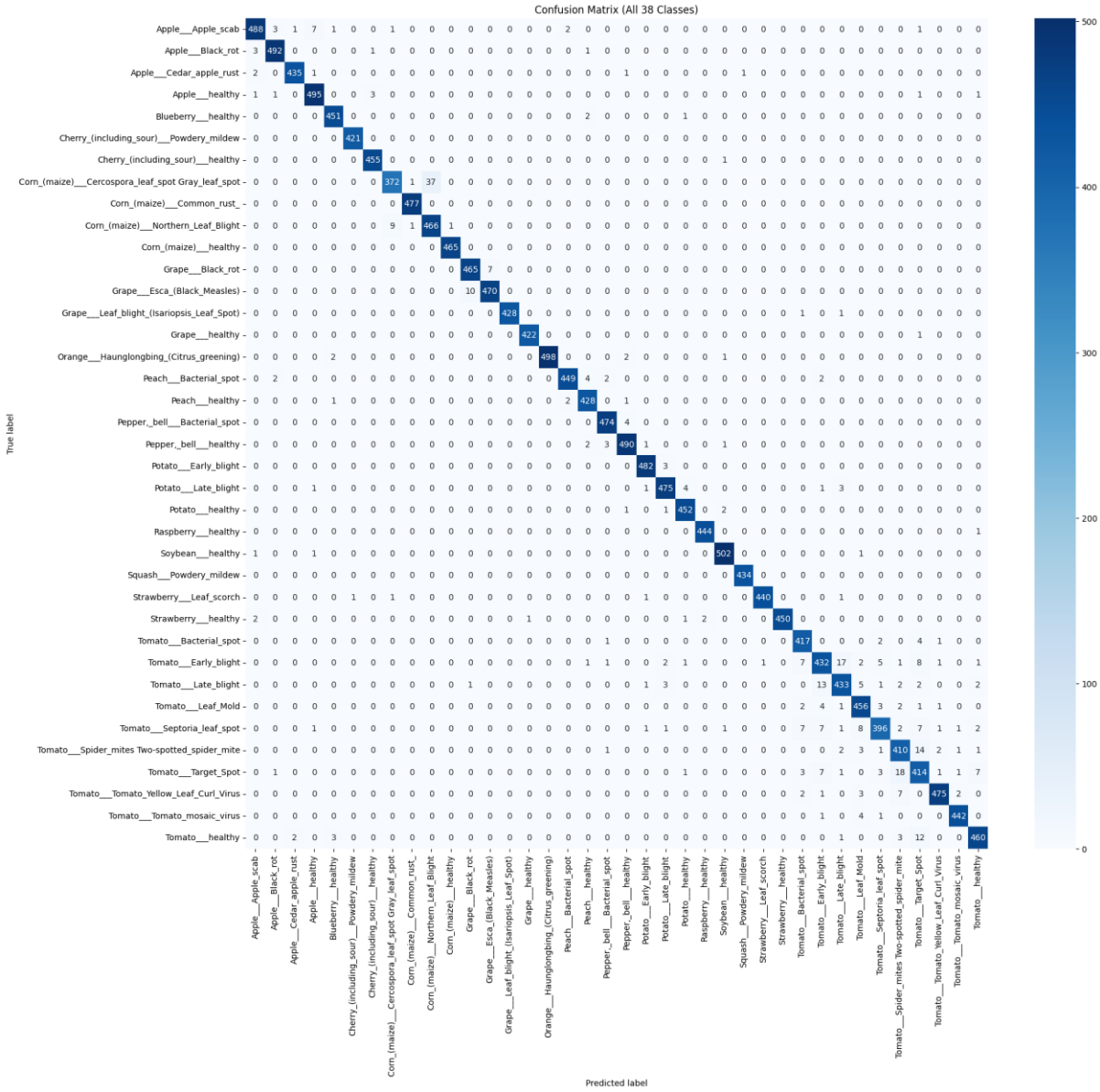


Figure 8: Confusion Matrix XGBOOST

Key points from Confusion Matrix:

1.Diagonal Elements (True Positives):

As shown in Figure 8,The diagonal elements correspond to number of times prediction was correct (i.e., prediction label = actual label).Most of the values on the diagonal are very high, which means that the model has great success in correctly recognizing the different classes of plant diseases. A few examples:

- Apple__Apple_scab: 468 correct predictions.
- Tomato__Tomato_mosaic_virus: 460 correct predictions.
- Grape__Esca_(Black_Measles): 470 correct predictions.

2.Off-Diagonal Elements (Misclassifications):

Off-diagonal elements describe those situations of misclassification in which there is heterogeneous class prediction and actual class**.**Several classes reveal greater numbers of misclassifications, which may imply that the model has difficulty differentiating the specific classes. For instance:Apple__Apple_scab: 7 misclassified instances are predicted as Apple__Black_rot and Apple__Cedar_apple_rust, thus confusion between these diseases is the most probable reason. Tomato_Early_blight: Misclassifications with Tomato__Late_blight.

3.High Accuracy for Some Classes:

- Several classes have **very high accuracy** with the correct predictions on the diagonal being consistently high. Some examples:
- **Peach__Bacterial_spot**: 492 correct predictions.
- **Pepper_bell__Bacterial_spot**: 479 correct predictions.

4.Performance Metrices XGBOOST

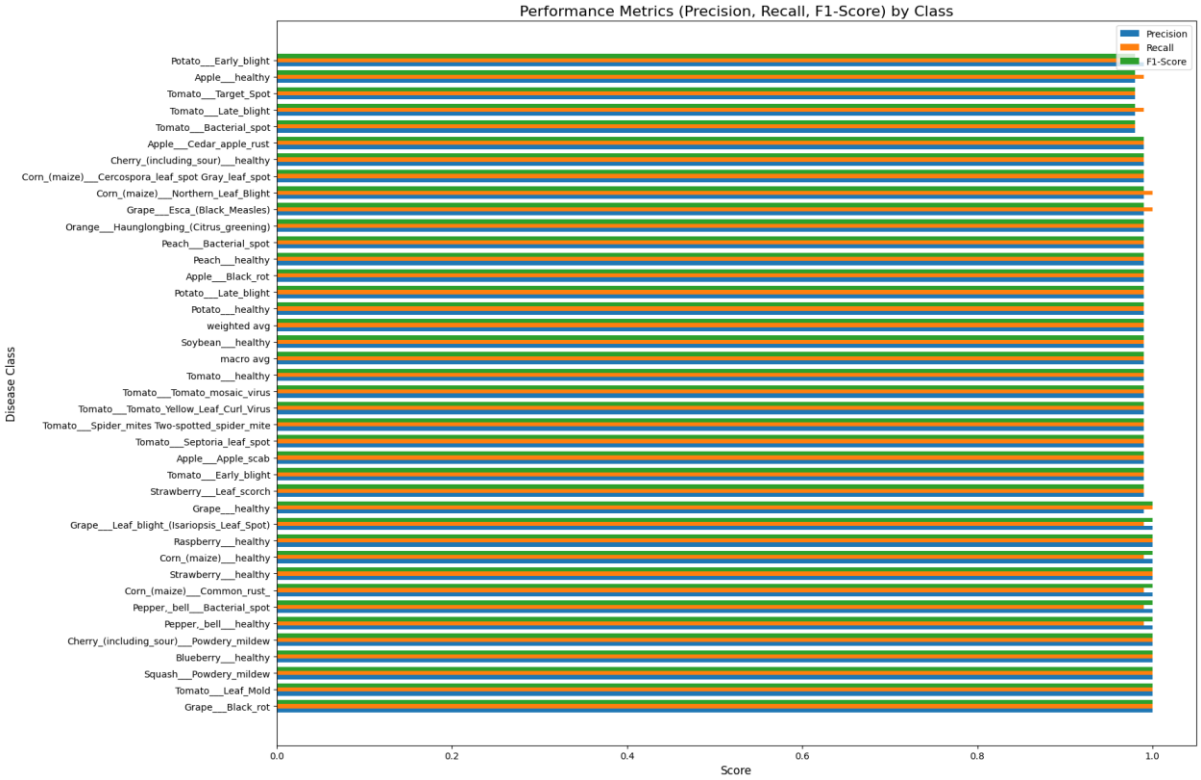


Figure 9: Performance Metrices XGBOOST

Overall Performance

As shown in Figure 9, high Performance across the Board: The model demonstrates very high performance for nearly all disease classes. The precision, recall, and F1-scores for most classes are extremely close to 1.0, indicating excellent classification accuracy.**F1-Score Dominance:** For almost every class, the green "F1-Score" bar completely overlaps the "Precision" and "Recall" bars, suggesting that the precision and recall values are nearly identical and very high. This is a strong indicator of a balanced and effective model.

Observations

Consistency: The model's performance is highly consistent across the vast majority of plant diseases shown. There is no single class that stands out as a significant weakness or a major outlier in terms of poor performance. “**Weighted avg” and "macro avg”:** The performance metrics for both the weighted and macro averages are also very high close to 1.0

Discussion

This research effectively demonstrates the potential of a **hybrid CNN-XGBoost model** for automated plant disease detection from leaf images. The model successfully addresses the core objective of building a high-performance system capable of accurately identifying diseases and healthy plants. A detailed analysis of the confusion matrices for both the CNN and XGBoost models provided valuable insights Though CNN exhibited some misclassifications, especially for classes "Class 6" and "Class 10," the hybrid model, in which the XGBoost component played a major role, was very successful in lowering these errors. In other words, the hybrid approach is very efficient in detail adjusting the classification and lessening the confusion that results from the diseases having a close visual aspect. The confusion matrix of the XGBoost model indicates that predictions most of the time are located along the diagonal, which is a confirmation of the model's accuracy in distinguishing different plant diseases.

The successful deployment of the model in a user-friendly application built with Streamlit. Not only does this app provide instant diagnostics, it also gives recommendations on fertilizers, care tips, and preventive measures, which is an extremely useful.

Conclusion

This research opens new avenues for implementing deep learning and hybrid machine learning models for identifying plant diseases. The usage of CNN and XGBoost has brought about the system towards achieving high classification accuracy, where the performance of CNN was 95% and the XGBoost classifier surpassed it at 97.63%. The transition from solely using CNN to combining XGBoost is well justified by the authors due to the fact that XGBoost could better balance the classification by reducing the number of misclassifications and improving the generalization over the different plant diseases besides being accurate in feature extraction from images.

Moreover, the Streamlit-based tool made by the researchers from the same study is an option that shows the pragmatic benefits of the investigation, the users can leaf images and the AI will give them precise disease diagnosis along with solutions and precautions to take care of the plant. This is an easy way to implement the technology in the absence of a direct line of sight between a research and a user, thus it automatically levels farmers, students, and other stakeholders in the agricultural community in terms of access to the latest equipment in the fight against plant diseases.

Finally, the suggested hybrid model not only increases the accuracy of prediction as desired but also ensures the accessibility to a real-world agricultural environment, thus the intelligent and data-driven solutions for sustainable agriculture and crop management have been enabled.

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